Individual plan

PROJECT INFORMATION

Title: eBPF-driven Fine-grained Telemetry and Machine Learning for EVPN/VXLAN Datacenter Networks

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1. BACKGROUND & OBJECTIVE

1.1 Background

EVPN/VXLAN-based Spine–Leaf architectures are widely used in modern datacenters to support scalable and flexible networking. However, short bursts of traffic, often called microbursts, can cause sudden queue buildup and lead to p99/p999 tail-latency spikes. These latency spikes can negatively affect latency-sensitive applications.

Most existing monitoring approaches detect congestion after it has already happened or provide only coarse-grained information. In contrast, eBPF allows fine-grained telemetry collection directly from the Linux kernel datapath without requiring special hardware. This creates an opportunity to study whether early signals of congestion can be observed and used for prediction.

1.2 Objective

The goal of this project is to explore whether kernel-level telemetry collected using eBPF can provide early signals of microburst-induced latency spikes, and to evaluate whether simple prediction-based mitigation can improve latency performance in a reproducible software-based testbed.

2. RESEARCH QUESTION & METHOD

2.1 Research Questions

1. Which kernel-level metrics collected via eBPF show early changes before latency spikes occur?

2. Can simple prediction models use these metrics to detect latency spikes shortly before they happen?

2.2 Hypothesis

1. Queue-related and burst-related kernel metrics contain early indicators of latency spikes.

2. Prediction-based mitigation can reduce tail latency without causing significant overhead.

2.3 Methods

Testbed Design

A small-scale, software-based testbed will be built using Linux network namespaces and FRR to emulate an EVPN/VXLAN Spine–Leaf network. Each Leaf node will connect to several host namespaces to represent tenant endpoints. This setup allows controlled traffic generation and reproducible experiments without requiring specialized hardware.

Microburst Reproduction

Microburst conditions will be reproduced using mixed traffic patterns, such as incast traffic and short-flow bursts. Traffic shaping and rate limiting will be used to create temporary congestion and observe its impact on latency.

Data Collection

Telemetry data will be collected using eBPF programs attached at the TC layer to capture metrics such as packet rate, drop counters, and queue-related indicators. End-to-end latency will be measured at the host level to provide ground truth for identifying latency spikes.

Data Processing & Labeling

The collected telemetry and latency data will be aggregated using sliding time windows. Latency spikes will be labeled based on deviations from baseline latency, enabling further analysis of potential predictive signals.

Early Feasibility Validation

A small proof-of-concept dataset will be generated in a minimal topology to demonstrate that latency spikes can be reproduced and aligned with kernel-level signals. This step will help verify the feasibility of the data collection and labeling process before extending the experiments to the full testbed.

2.4 Limitations

This study has several limitations. First, the experiments will be conducted in a small-scale software-based testbed, which may not fully capture the complexity of production datacenter networks. Second, the telemetry collected through eBPF is limited to observable kernel-level signals and may not reflect all factors affecting latency. Third, the prediction models in this project will remain relatively simple, so the results may mainly demonstrate feasibility rather than optimal prediction accuracy. Finally, the proposed mitigation strategies will be evaluated only in a controlled environment, and their effectiveness in large-scale real-world deployments will require further study.

3. EVALUATION & NEWS VALUE

3.1 Evaluation

The success of the project will be evaluated by determining whether the proposed approach can identify and predict latency spikes and whether prediction-based mitigation can improve performance.

Both quantitative and qualitative measures will be used in this section.

Quantitative metrics includes: p99 and p999 end-to-end latency, latency spike frequency and duration, packet loss rate, throughput under different traffic conditions, CPU overhead introduced by telemetry and mitigation

Qualitative evaluation includes: Observing whether kernel-level telemetry shows consistent patterns before latency spikes. Assessing the practicality of deploying software-based telemetry in a controlled environment

3.2 Expected Scientific Results

This work is scientifically relevant because it explores predictive observability in datacenter networks using software-based telemetry. The study aims to improve understanding of how kernel-level metrics relate to transient congestion events and tail-latency behavior.

The results may provide insights into: the relationship between queue dynamics and latency spikes**.** The feasibility of using eBPF for fine-grained network observability**.** The effectiveness of lightweight mitigation strategies

3.3 News value

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4. PRE-STUDY

The pre-study will focus on key topics related to datacenter network performance and observability.

First, microburst behavior and its impact on queue buildup and latency will be reviewed, as these transient events are a known cause of tail-latency spikes in datacenter networks [1].

Second, existing work on tail latency will be studied to understand its importance as a performance metric and how it is affected by network congestion [2].

Third, network telemetry techniques will be examined, including both hardware-based approaches such as in-band network telemetry and software-based approaches using eBPF, in order to understand how fine-grained network measurements can be obtained [3], [4].

Finally, prior work on congestion detection and mitigation in datacenter networks will be reviewed to identify existing approaches and their limitations, especially in handling short-lived traffic bursts [5].

These areas will provide the necessary background and help position the proposed work within current research.

5. CONDITIONS & SCHEDULE

5.1 Required Resource

Hardware: A personal workstation or server capable of running multiple network namespaces and optional access to university lab resources if available

Software: Linux operating system, FRR for EVPN/BGP control plane, iproute2 for network configuration, eBPF toolchain (e.g., bpftool, libbpf, or BCC), traffic generation tools (e.g., iperf3), Python for data processing and analysis

Data: Experimentally generated traffic data from the testbed

5.2 Supervisor Involvement

The supervisor will provide guidance on research scope, methodology, and evaluation criteria. Regular meetings will be held to discuss progress, challenges, and adjustments to the project plan.

5.3 Project Timeline

Phase1: Environment and Data Collection (Apr~May)

Goals

Build EVPN/VXLAN testbed, generate a POC dataset, and verify latency spike reproduction and signal alignment

Deliverables

POC dataset, latency and queue signal plots, verified data collection pipeline

Phase2: Feature Analysis and Prediction (Jun~Jul)

Goals

Build a labeled dataset, analyze relationships between features and spikes, and train and evaluate prediction models

Deliverables

Feature analysis results, prediction performance metrics

Phase3: Mitigation Strategies and System Evaluation (Sep~Oct)

Goals: Implement simple mitigation strategies

Compare baseline and prediction-based approaches, and evaluate latency improvements

Deliverables

Performance comparison results, overhead analysis, and final conclusions

USE OF GENERATIVE AI

chatGPT5.2 is used in this individual plan for following contents: structure designing, limitation analysis, schedule making.

DeepL is used for translation.

REFERENCES

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[5] M. Ghorbani et al., “DRILL: Micro load balancing for low-latency data center networks,” in Proc. ACM SIGCOMM, 2017.